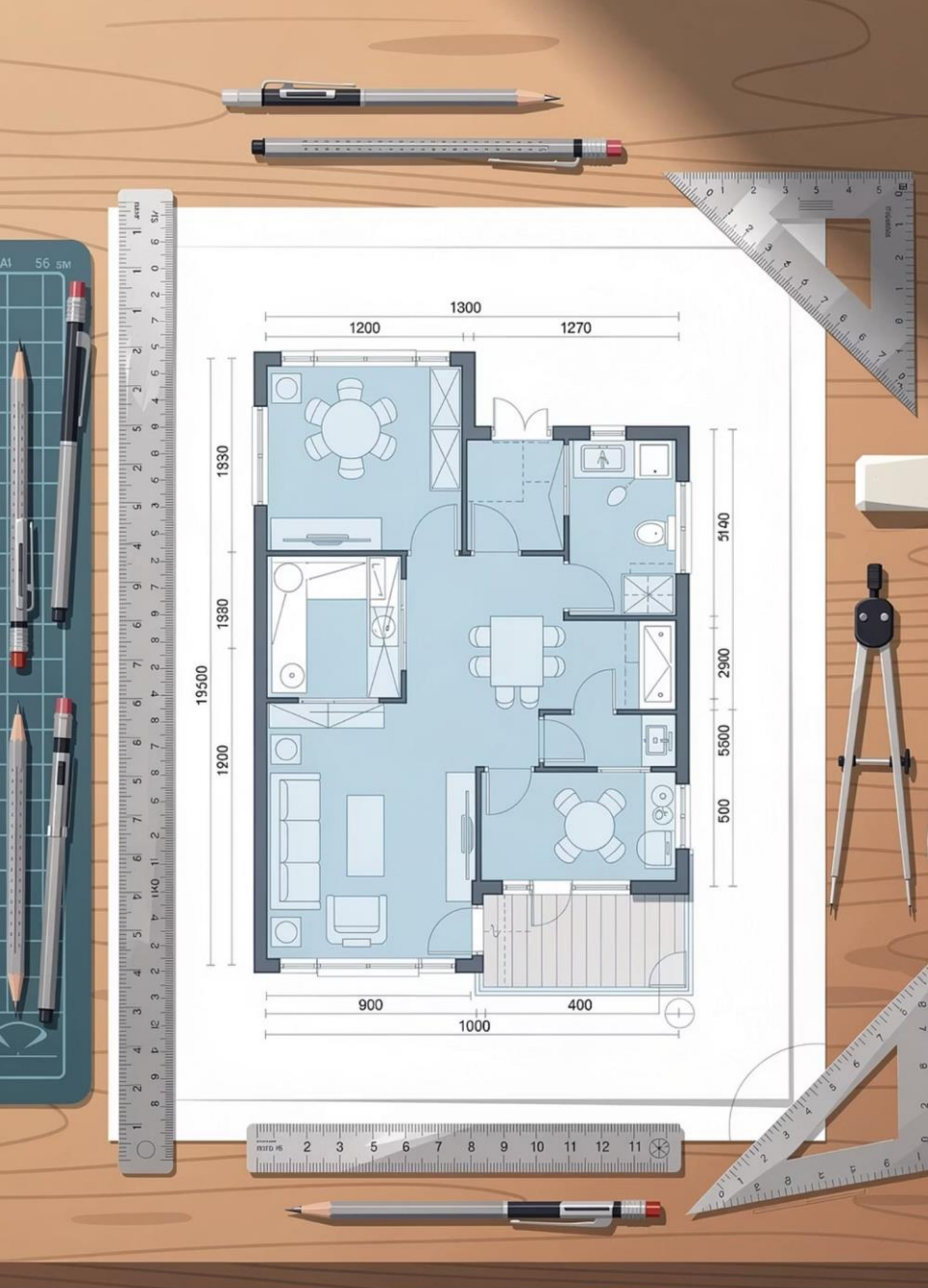




Cloud Architecture & The AWS Well-Architected Framework

ITNW 1009 · Fundamentals of Cloud and AI

Individual AWS services are the bricks. **Cloud architecture is the blueprint.**

Today we step into the role of the Cloud Architect – learning to connect services strategically, evaluate design decisions against proven best practices, and build systems that are resilient, secure, efficient, and cost-conscious.

What We're Covering



The Well-Architected Framework

AWS's six-pillar master checklist for evaluating cloud architectures.



Disaster Recovery Models

RPO, RTO, and the four AWS DR strategies from cheapest to fastest.



High Availability vs. Fault Tolerance

Understanding the critical trade-off between uptime strategies and cost.



AWS Trusted Advisor

Your automated cloud consultant — continuous auditing across five domains.

What Is Cloud Architecture?

Physical Building Analogy

You wouldn't throw bricks, pipes, and electrical wires into a pile and hope it becomes a sturdy building. You need an architectural blueprint that tells you exactly how every component connects, supports, and protects the others.

Cloud architecture works the same way — it's the **strategic discipline** of combining AWS services into flexible, resilient systems designed to solve real business problems safely and efficiently.

Why Architecture Decisions Matter

- **Performance:** Right-sized resources serving users at speed
- **Security:** Protecting data, identities, and network boundaries
- **Cost:** Eliminating waste and idle infrastructure
- **Reliability:** Recovering automatically from failures
- **Sustainability:** Reducing environmental impact of workloads

Every app crash during a major sale event traces back to an architectural decision someone made — or forgot to make.



The AWS Well-Architected Framework

Built From Thousands of Real-World Customer Reviews

AWS developed this formal framework by reviewing real customer architectures at massive scale — identifying every failure mode and success pattern. The result is a **master checklist organized around six core pillars**. Think of it as a temple: if even one pillar is weak, the entire roof — your business application — risks collapse. All six pillars must be strong and balanced simultaneously.

The Six Pillars

Operational Excellence

Running, monitoring, and improving systems continuously.

Sustainability

Minimizing the environmental impact of cloud workloads.

Cost Optimization

Delivering maximum value at the lowest responsible price point.



Security

Protecting data, systems, and assets at every layer.

Reliability

Recovering from failures automatically and scaling to meet demand.

Performance Efficiency

Using the right resources efficiently as demand and technology evolve.

Pillars 1 & 2: Operational Excellence & Security

Operational Excellence

In traditional IT, admins updated servers manually. One typo could break production. In the cloud, **Operational Excellence means treating operations as code** — automating deployments, monitoring every change, and designing systems to recover automatically from small errors.

- Automate deployments via CI/CD pipelines
- Make frequent, small, reversible changes
- Learn from every failure — post-mortems are standard
- Anticipate failure modes in advance and design around them

Security

Security in the cloud means applying the **Principle of Least Privilege** — giving people and systems only the exact access they need. Nothing more. It also means encrypting data in transit and at rest, enabling traceability for every action, and automating security responses.

- IAM: Least privilege access control
- Encrypt all data at rest and in transit
- Enable multi-factor authentication (MFA)
- Maintain audit trails for all administrative actions

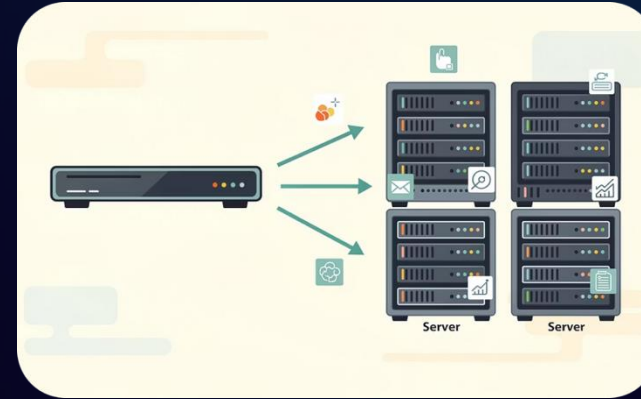
Pillars 3 & 4: Reliability & Performance Efficiency



Reliability

A reliable system **assumes things will break** — and is designed to recover automatically. Instead of one massive server, distribute smaller instances across multiple Availability Zones. If Zone A loses a server, Zone B instantly absorbs the traffic. Key practices:

- Auto Scaling Groups across multi-AZ deployments
- Automated failure detection and recovery
- Test recovery procedures regularly



Performance Efficiency

Pick the right tool for the job and **use cloud resources efficiently**. Don't run an oversized database server if a serverless key-value store handles the load faster and cheaper. Place assets close to users via CloudFront Edge Locations. Key practices:

- Go serverless where possible (Lambda, DynamoDB)
- Use ElastiCache for read-heavy caching layers
- Leverage CloudFront for global static asset delivery

Pillars 5 & 6: Cost Optimization & Sustainability

Cost Optimization

Cloud computing converts heavy upfront capital expenses (CapEx) into flexible operational expenses (OpEx). But leaving idle resources running destroys budgets. **Optimization means paying only for what you consume.**

- Adopt a consumption model — shut down idle assets
- Use Reserved Instances or Savings Plans for predictable workloads
- Eliminate unattached EBS volumes, Elastic IPs, unused Load Balancers
- Right-size instances based on actual utilization metrics

Sustainability

The newest pillar — and a growing responsibility. Running data centers requires enormous electricity and cooling. Cloud architects have an ethical duty to reduce environmental impact.

- Maximize hardware utilization — fewer, higher-utilized instances
- Use S3 lifecycle rules to move cold data to low-energy Glacier storage
- Choose efficient compute families (e.g., Graviton processors)
- Reducing idle compute directly supports both Cost AND Sustainability simultaneously

- ✓ Stopping an unattached EBS volume saves money on your bill AND stops wasting physical storage infrastructure energy — a perfect double win across two pillars.

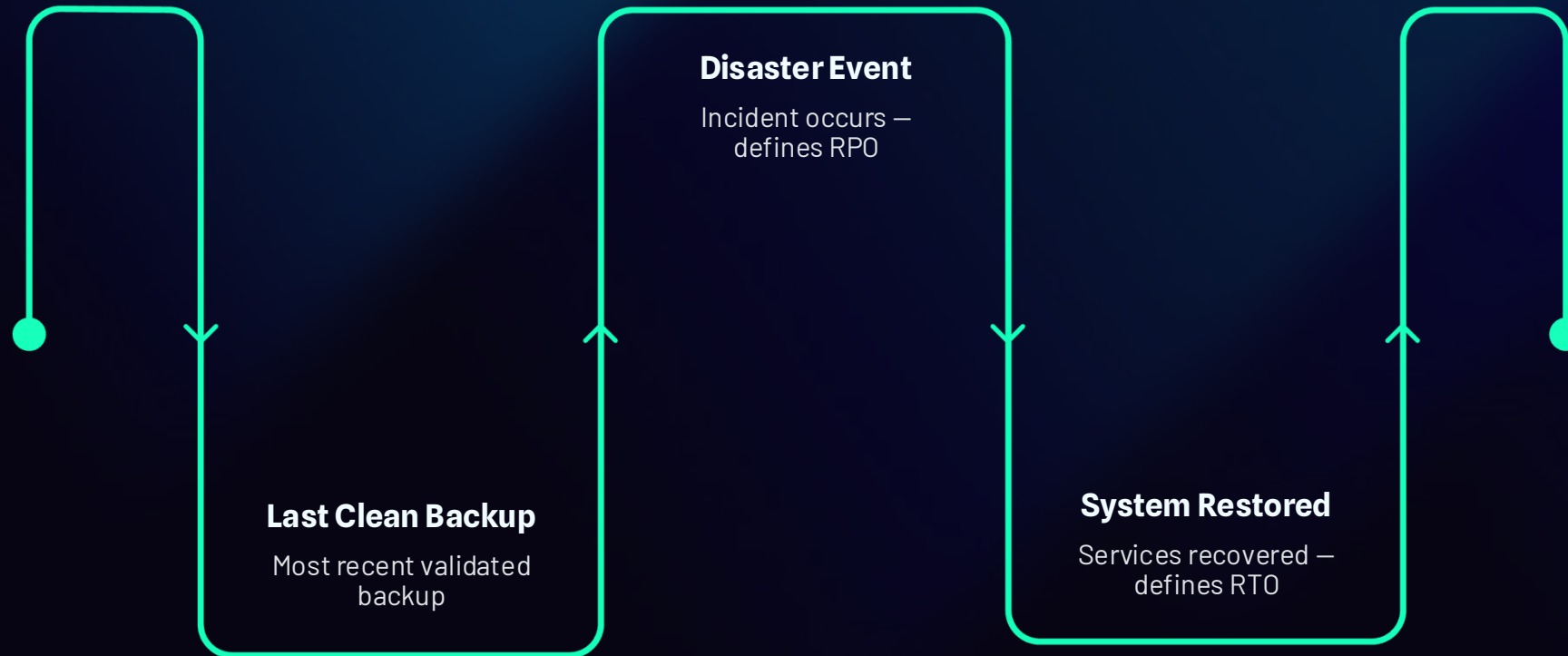
High Availability vs. Fault Tolerance

Understanding the Trade-Off

Dimension	High Availability (HA)	Fault Tolerance (FT)
Definition	System remains accessible with minimal downtime; brief failover micro-pauses acceptable.	Zero downtime and zero loss of operational state – even if hardware fails mid-transaction.
Downtime Target	99.99% uptime – allows seconds of failover recovery.	100% uptime – absolutely zero service interruption or state loss.
Architecture	Multi-AZ with ELB + Auto Scaling; secondary instance stands by.	Full active-active redundancy – duplicate hardware running in parallel at all times.
Cost	Cost-effective – meets 99% of business requirements.	Significantly more expensive – requires complete hardware duplication.

A hospital's emergency life-support network requires **Fault Tolerance**. An online clothing retail store thrives on **High Availability**. Know your business requirements before choosing your design model.

Disaster Recovery: RPO & RTO



RPO — Recovery Point Objective

How much data can we afford to lose? If you back up nightly at midnight and your system crashes at 11 PM, you just lost 23 hours of data. If your RPO is 5 minutes, you need continuous database replication – not nightly snapshots.

RTO — Recovery Time Objective

How long can we afford to be offline? Can your business tolerate a 4-hour restore window, or do you need to be back online in 30 seconds? Higher RTO tolerance = lower infrastructure cost. Lower RTO = active redundancy required.

The Four AWS Disaster Recovery Strategies



Backup & Restore

Cheapest option. Periodic backups stored in S3. Highest RTO/RPO — slow to recover. Great for non-critical workloads.



Pilot Light

Core data continuously replicated to secondary Region. Infrastructure offline until disaster strikes — then rapidly scaled up. Lower cost than warm standby.



Warm Standby

Scaled-down version of the full production stack running continuously in a secondary Region. Faster recovery than Pilot Light — scale up at failure time.



Multi-Site Active-Active

Full production traffic handled in real-time across multiple Regions simultaneously. Near-zero RTO and RPO. Most expensive — reserved for mission-critical systems.



AWS Trusted Advisor

Your Automated Cloud Consultant

AWS Trusted Advisor continuously scans your entire AWS environment and compares your configurations against Well-Architected best practices — providing **real-time, color-coded recommendations** across five critical domains. Think of it as a personal security consultant and accountant living inside your management console, alerting you before problems become crises.

Trusted Advisor — Five Domains



Cost Optimization

Flags idle resources – unattached EBS volumes, unused Elastic IPs, underutilized Reserved Instances. Every yellow flag is money walking out the door.



Performance

Identifies overloaded EC2 instances, unoptimized CloudFront configurations, and bottlenecked database connections affecting user experience.



Security

Red alerts for open S3 buckets, unrestricted SSH access, MFA disabled on root accounts, and publicly exposed security group ports.



Fault Tolerance

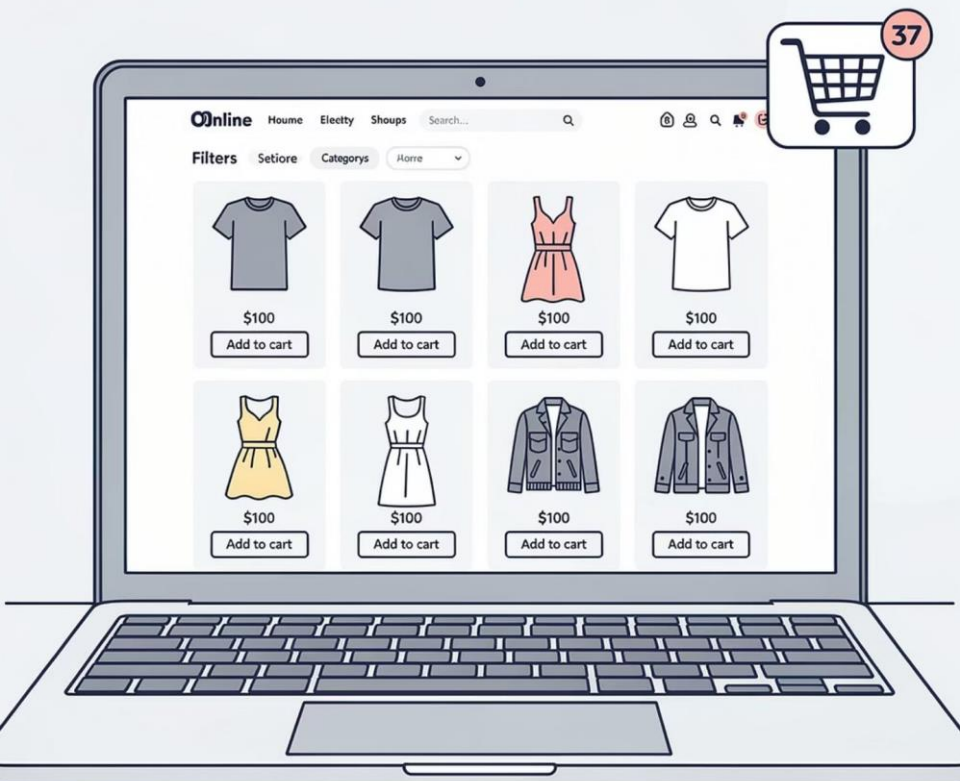
Warns on Single-AZ RDS instances, missing Auto Scaling groups, and EC2 instances without multi-AZ deployment configurations.



Service Limits

Alerts when you approach 80–100% of AWS service quotas (EC2 limits, VPC limits, EIP allocations) – before your next deployment fails.

Status indicators: **Green**  = No action needed · **Yellow**  = Investigate · **Red**  = Act immediately.



Active Learning — Architecture Workshop

Apex Retail — Audit & Re-Architect

You've been hired as Junior Cloud Architects by **Apex Retail**. During their last holiday flash sale, their web application suffered a **4-hour outage costing \$150,000** in lost sales. You are reviewing their current infrastructure and it's... not good. Time to fix it using the Well-Architected Framework.



Format: Think-Pair-Share — 5 min individual analysis → 10 min pair discussion → 10 min instructor debrief. Use your worksheet to map pillar violations before discussing.

Apex Retail — Identified Flaws

Flaw 1: Single EC2 Instance

The entire web application runs on **one EC2 instance** in a single public subnet. No load balancer, no Auto Scaling — one failure and the whole application goes dark. This violates **Reliability**.

Flaw 2: Images on Local EBS

Product catalog images uploaded by users are saved directly onto the **local EBS volume** of that single EC2 instance. If the instance fails, those images are gone — and serving them from compute saturates disk I/O. Violates **Performance Efficiency**.

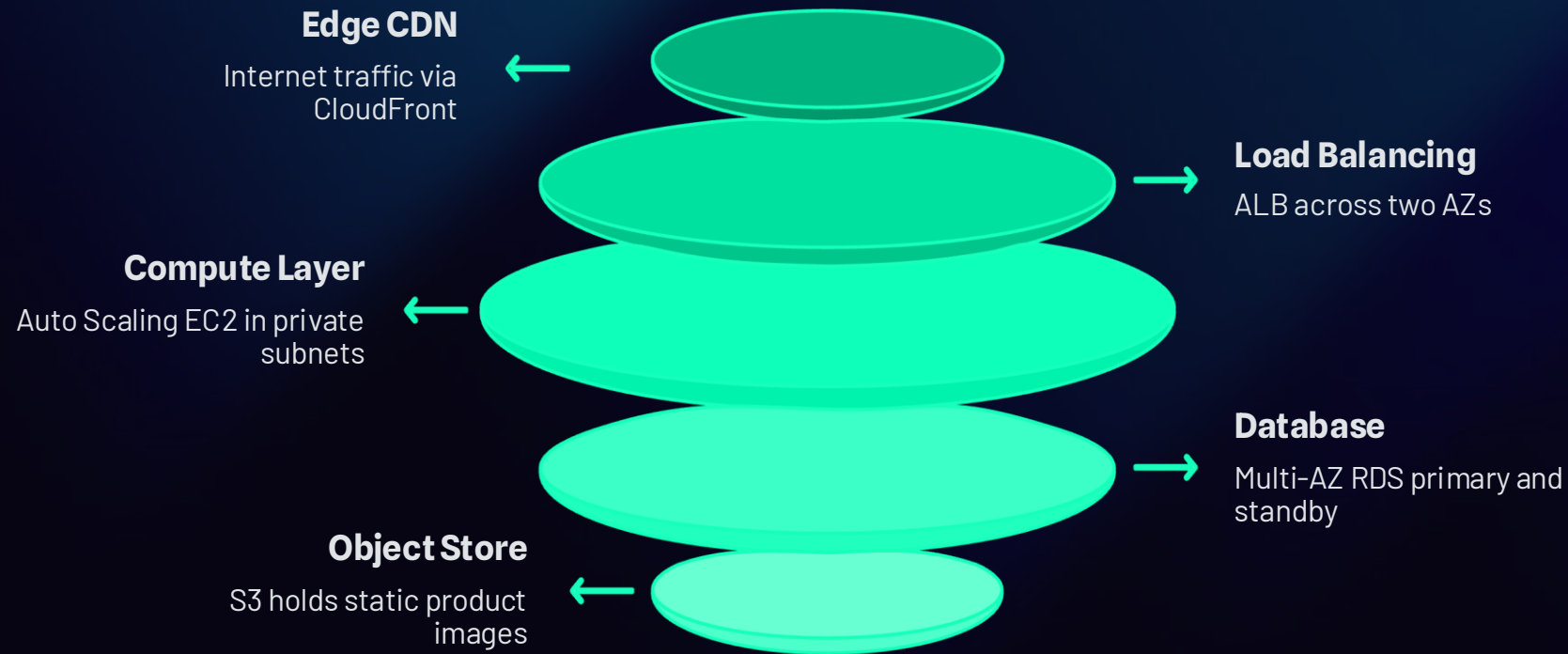
Flaw 3: Single-AZ RDS, No Backups

The MySQL database runs in a **Single Availability Zone with automated backups disabled** to save \$15 per month. The \$150,000 outage cost makes this the most expensive \$15 ever saved. Violates **Reliability** and **Cost Optimization**.

Flaw 4: Manual SSH Deployments

Developers deploy code by **SSH-ing into the live production server** and manually replacing files during peak shopping hours. One typo = outage. Violates **Operational Excellence** and **Security**.

Apex Retail — Re-Architected Solution



ALB + Multi-AZ ASG

Application Load Balancer in public subnets distributes traffic to EC2 servers spread across two private Availability Zones — no single point of failure.

S3 + CloudFront

Product images moved to Amazon S3, served globally via CloudFront CDN — eliminating local disk I/O bottlenecks and reducing server load dramatically.

Multi-AZ RDS

MySQL upgraded to Multi-AZ RDS with automated daily snapshots. Synchronous replication to a standby replica — automatic failover in 60–120 seconds.

The Restaurant & Airline Analogies

Making Well-Architected Stick

Commercial Restaurant

- **Well-Architected Framework** → Master Restaurant Management Standard
- **High Availability** → Multiple cooking line stations; one cook steps away, another takes over
- **Fault Tolerance** → Dual gas lines + automatic backup generators — zero appliances ever shut off
- **RTO** → Manager's target: pipe leak cleaned, guests reseated within 15 minutes
- **RPO** → Carbon-copy ticket pad: only lose 5 minutes of orders if the screen crashes
- **Trusted Advisor** → Surprise health & financial auditor with a color-coded checklist

Commercial Airline

- **Well-Architected Framework** → FAA Aviation Safety & Ops Standards
- **High Availability** → Standby aircraft at the adjacent gate — 45-minute transfer, flight reaches destination
- **Fault Tolerance** → Redundant engines, independent hydraulic channels — Engine 1 fails, Engine 2 sustains full altitude
- **RTO** → Ground maintenance turnaround: tire swapped, plane certified in 30 minutes
- **RPO** → Satellite-synced manifests: passenger records lost only up to 1 minute before crash
- **Trusted Advisor** → Automated telemetry scanner flagging worn tires (yellow) or uncalibrated radar (red)

Key Takeaways

Six Pillars, Balanced Always

A system that's secure but costs \$100K/month when revenue is \$10K is not well-architected. Every pillar must be evaluated — none can be ignored.

HA vs. FT Is a Business Decision

High Availability meets 99% of business needs at reasonable cost. Fault Tolerance is for mission-critical systems where even milliseconds of downtime cause harm.

RPO and RTO Define Your DR Strategy

Know how much data loss and how much downtime your business can tolerate — then choose the DR strategy (Backup, Pilot Light, Warm Standby, or Active-Active) that matches.

Trusted Advisor Is Always Watching

Use Trusted Advisor proactively — don't wait for an outage or a surprise bill to discover vulnerabilities. Green means go; red means act now.